

GCP Hands-On:

Hands-on experience of core Google Cloud features.

Event-Driven Messaging with Pub/Sub:

Setup:

```
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.  
shankaraditya75@cloudshell:~ (gcp-hands-on-490407) $ gcloud services enable pubsub.googleapis.com  
Operation "operations/acf.p2-130288429366-301a2947-8b2a-4a0a-a09d-bc2f345a2d36" finished successfully.  
shankaraditya75@cloudshell:~ (gcp-hands-on-490407) $
```

Publisher & Subscriber:

Creation:

```
$ gcloud pubsub topics create my-first-topic  
Created topic [projects/gcp-hands-on-490407/topics/my-first-topic].
```

Confirmation:

```
$ gcloud pubsub topics list  
gcloud pubsub subscriptions list  
---  
name: projects/gcp-hands-on-490407/topics/my-first-topic  
---  
ackDeadlineSeconds: 10  
expirationPolicy:  
  ttl: 2678400s  
messageRetentionDuration: 604800s  
name: projects/gcp-hands-on-490407/subscriptions/my-sub  
pushConfig: {}  
state: ACTIVE  
topic: projects/gcp-hands-on-490407/topics/my-first-topic
```

```
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$ gcloud pubsub topics create my-first-topic
Created topic [projects/gcp-hands-on-490407/topics/my-first-topic].
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$ gcloud pubsub subscriptions create my-sub --topic=my-first-topic
Created subscription [projects/gcp-hands-on-490407/subscriptions/my-sub].
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$ gcloud pubsub topics list
gcloud pubsub subscriptions list
---
name: projects/gcp-hands-on-490407/topics/my-first-topic
---
ackDeadlineSeconds: 10
expirationPolicy:
  ttl: 2678400s
messageRetentionDuration: 604800s
name: projects/gcp-hands-on-490407/subscriptions/my-sub
pushConfig: {}
state: ACTIVE
topic: projects/gcp-hands-on-490407/topics/my-first-topic
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$
```

Hello World Message:

```
$ gcloud pubsub topics publish my-first-topic --message="Hello from the terminal!"
```

```
messageIds:
```

```
- '18043918666663806'
```

```
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$ gcloud pubsub topics publish my-first-topic --message="Hello from the terminal!"
messageIds:
- '18043918666663806'
shankaraditya75@cloudshell:~ (gcp-hands-on-490407)$ gcloud pubsub subscriptions pull my-sub --auto-ack
DATA: Hello from the terminal!
MESSAGE_ID: 18043918666663806
ORDERING_KEY:
ATTRIBUTES:
DELIVERY_ATTEMPT:
ACK_STATUS: SUCCESS
```

Deletion:

```
$ gcloud pubsub subscriptions delete my-sub
```

```
gcloud pubsub topics delete my-first-topic
```

```
Deleted subscription
```

```
[projects/gcp-hands-on-490407/subscriptions/my-sub].
```

```
Deleted topic [projects/gcp-hands-on-490407/topics/my-first-topic].
```

Conclusion:

This hands-on exercise demonstrated the implementation of an **asynchronous messaging pattern**, a cornerstone of resilient and scalable cloud-native applications. By moving away from direct point-to-point communication, I explored how services can remain decoupled while maintaining high data throughput.

Key Takeaways:

- **Asynchronous Decoupling:** Successfully separated the **Publisher** (the message source) from the **Subscriber** (the message consumer). This architecture ensures

that the sender does not need to wait for the receiver to be active, allowing for greater system flexibility and fault tolerance.

- **Topic-Subscription Relationship:** Configured a **1-to-N relationship** where a single topic can broadcast to multiple subscriptions. I practiced the lifecycle of a message—from ingestion into a topic to persistent storage in a subscription queue until it is manually acknowledged.
- **CLI-Driven Orchestration:** Demonstrated proficiency with the `gcloud pubsub` toolset to manage resources without the overhead of a GUI. This included service enablement, topic creation, and the programmatic publishing/pulling of data packets.
- **Stateless Scaling:** Gained insight into how Pub/Sub handles the "burstiness" of data. In a real-world **Data Engineering** pipeline, this setup would act as a buffer between a high-traffic web app and a backend database like Bigtable or BigQuery.
- **Cleanup and Governance:** Followed the "SRE Lifecycle" by immediately decommissioning the topic and subscription after the test, demonstrating responsible resource management and cost awareness.